# Statistics, Counting & Probability

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**ANSWER KEY — COMPLETE WORKED SOLUTIONS**

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A box contains 5 red and 3 blue balls. Find the probability that 3 balls drawn without replacement are all red.

Method 1 (Sequential Conditional Probability): Total balls = $5 + 3 = 8$. 1st ball red: $5/8$ 2nd ball red: $4/7$ 3rd ball red: $3/6 = 1/2$ $P = \frac{5}{8} \times \frac{4}{7} \times \frac{3}{6} = \frac{60}{336} = \mathbf{\frac{5}{28}} \approx 0.1786$ Method 2 (Combinations): $P = \frac{\binom{5}{3}}{\binom{8}{3}} = \frac{10}{56} = \mathbf{\frac{5}{28}}$

**2** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

If $P(A) = 0.6$ and $P(B|A) = 0.4$, find the joint probability $P(A \cap B)$.

Apply the conditional probability multiplication rule $P(A \cap B) = P(B \mid A) \cdot P(A)$: $P(A \cap B) = 0.4 \times 0.6 = \mathbf{0.24}$ Key Mathematical Principle & Verification: In exam revision, always verify that the final value satisfies the domain constraints, units, and original governing equation by direct back-substitution into the prompt.

**3** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A box contains 3 red, 4 white, and 5 blue cards. Find the probability of drawing a card that is not white.

Step 1: Total cards: $3 + 4 + 5 = 12$. Step 2: Non-white cards: $3 \text{ red} + 5 \text{ blue} = 8$ (or $12 - 4 = 8$). Step 3: Probability: $P(\text{not white}) = \frac{8}{12} = \mathbf{\frac{2}{3}} \approx 0.667$

**4** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the vertical asymptote(s) of $f(x) = \frac{3x}{x^2 - 4}$.

Step 1: Vertical asymptotes occur where the denominator is zero while the numerator is non-zero. $x^2 - 4 = 0 \Rightarrow (x - 2)(x + 2) = 0 \Rightarrow x = 2$ and $x = -2$ Step 2: Verify numerator at these values: $3(2) = 6eq0$ and $3(-2) = -6eq0$. The vertical asymptotes are the lines $\mathbf{x = 2}$ and $\mathbf{x = -2}$.

**5** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set has standard deviation 3 and mean 20. If every value in the data set is increased by 5, find the new mean and new standard deviation.

New Mean: Adding a constant $c = 5$ shifts the mean by $+5$: $\text{New Mean} = 20 + 5 = \mathbf{25}$ New Standard Deviation: Standard deviation measures dispersion relative to the mean; adding a constant shifts all points uniformly without altering spread: $\text{New Standard Deviation} = \mathbf{3}$ (unchanged)

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**6** [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Evaluate the finite series $\sum_{k=1}^{4}(2k-1)$.

Step 1: Expand the summation terms for $k = 1, 2, 3, 4$: $k = 1 : 2(1) - 1 = 1$ $k = 2 : 2(2) - 1 = 3$ $k = 3 : 2(3) - 1 = 5$ $k = 4 : 2(4) - 1 = 7$ Step 2: Add the terms: $1 + 3 + 5 + 7 = \mathbf{16}$ (Property: The sum of the first $n$ odd positive integers is $n^2 = 4^2 = 16$).

**7** [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Expand $(a + b)^5$ and find the coefficient of the term $a^2b^3$.

By the Binomial Theorem, the general term in $(a + b)^n$ is $T = \binom{n}{k}a^{n-k}b^k$. For $a^2b^3$, $n = 5$ and $k = 3$: $\text{Coefficient} = \binom{5}{3} = \frac{5!}{3! \cdot 2!} = \frac{5 \times 4}{2 \times 1} = \mathbf{10}$

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## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The mean of 10 numbers is 15. One number is removed and the new mean of the remaining 9 numbers is 14. Find the removed number.

Step 1: Total sum of the original 10 numbers: $S_{10} = 10 \times 15 = 150$ Step 2: Total sum of the remaining 9 numbers: $S_9 = 9 \times 14 = 126$ Step 3: The removed value is: $x = S_{10} - S_9 = 150 - 126 = \mathbf{24}$

**9** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the amplitude and period of $y = -4\cos\left(\frac{\pi x}{3}\right)$.

Amplitude: $|A| = |-4| = \mathbf{4}$. Period: For $y = A\cos(Bx)$ with $B = \frac{\pi}{3}$: $T = \frac{2\pi}{|B|} = \frac{2\pi}{\pi/3} = 2\pi \times \frac{3}{\pi} = \mathbf{6}$

**10** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the distance from $(1,\ -2)$ to $(5,\ 1)$.

Apply the Euclidean distance formula $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$:

$d = \sqrt{(5-1)^2 + (1-(-2))^2} = \sqrt{4^2 + 3^2} = \sqrt{16 + 9} = \sqrt{25} = \mathbf{5}$

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**11** **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

The bar graph below shows the distribution of customer ratings given to a local bakery by 20 customers on a scale from 1 to 5 stars. How many additional 5-star ratings must the bakery receive so that the overall mean rating of all customers becomes exactly 3.5 stars? Show your complete algebraic work.

Final Answer: 8. First, calculate the current sum of the ratings from the bar graph: $\text{Sum} = (1 \times 2) + (2 \times 6) + (3 \times 6) + (4 \times 4) + (5 \times 2) = 2 + 12 + 18 + 16 + 10 = 58$ The current number of customers is $2 + 6 + 6 + 4 + 2 = 20$. Let $x$ represent the number of additional 5-star ratings. Adding $x$ ratings of 5 stars increases the total sum of ratings by $5x$ and increases the total number of customers to $20 + x$. Set up the equation for the new mean.

**12** **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

The bar graph below shows the number of students enrolled in each of four elective classes at a high school. Each student is enrolled in exactly one of these four classes. If one student is selected at random from these four classes, what is the probability that the student is enrolled in Drama? Show your complete algebraic work.

Final Answer: $\frac{4}{25}$. First, find the total number of students enrolled in the four classes by adding the frequencies from the bar graph: $\text{Total} = 12 + 16 + 8 + 14 = 50$ From the graph, the number of students enrolled in Drama is $8$. The probability of selecting a student enrolled in Drama is the ratio of the number of Drama students to the total number of students: $P(\text{Drama}) = \frac{8}{50} = \frac{4}{25}$ - $\frac{1}{4}$ incorrectly assumes each elective has an equal number of students..

**13** **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

A committee of 4 people is to be selected at random from a group of 10 candidates consisting of 4 mathematicians, 3 physicists, and 3 engineers. Given that the selected committee contains at least one person from each of the three professions, what is the probability that exactly 2 mathematicians are on the committee? Show your complete algebraic work.

Final Answer: $\frac{3}{7}$. To find the conditional probability, we determine the number of committee selections that satisfy the given condition (the reduced sample space) and the number of those selections that contain exactly 2 mathematicians. 1. **Analyze the condition:** The committee must have 4 members chosen from 3 professions with at least one member from each profession. By the Pigeonhole Principle, the members must be partitioned as $2 + 1 + 1$. Thus, exactly one profession will have 2 representatives.

**14** **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Of 31 students, 13 play soccer and 11 of those also play basketball. What is the probability that a randomly chosen soccer player also plays basketball? Show your complete algebraic work.

Final Answer: $\frac{11}{13}$. Conditional probability restricts to the soccer group: $P(\text{basketball} \mid \text{soccer}) = \frac{11}{13} = \frac{11}{13}$.

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## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 1.01 Advanced Number Theory**

A school committee of 4 members is to be selected from a club consisting of 6 juniors and 5 seniors. (a) In how many total ways can the 4 members be selected? (b) In how many ways can the committee be formed if it must contain exactly 2 juniors and 2 seniors?

(a) Unrestricted selection: Total students $= 6 + 5 = 11$. Choosing 4 where order does not matter: $\binom{11}{4} = \frac{11 \times 10 \times 9 \times 8}{4 \times 3 \times 2 \times 1} = 11 \times 10 \times 3 = \mathbf{330}$ ways. (b) Exactly 2 juniors and 2 seniors: Select 2 juniors from 6, AND 2 seniors from 5: $\binom{6}{2} \times \binom{5}{2} = \binom{6 \times 5}{2} \times \binom{5 \times 4}{2} = 15 \times 10 = \mathbf{150}$ ways.

**16** **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)** • **Cross-Lesson: 2.01 Advanced Algebra & Systems**

A game pays $x15if a player rolls a 6 on a fair six − sided die, pays \6$ if they roll a 5, and the player must pay $x3$ for any other roll (1, 2, 3, or 4). Determine the expected gain or loss per roll, and state whether the game is favorable to the player.

Step 1 (Probability Breakdown): $P(\text{roll } 6) = 1/6$ with gain $x15$. P(\text{roll } 5) = 1/6 with gain +\6. $P(\text{roll } 1, 2, 3, \text{ or } 4) = 4/6$ with loss $x3$. $Step 2 (Expected Value)$ : E(X) = 15\left(\frac{1}{6}\right) + 6\left(\frac{1}{6}\right) + (-3)\left(\frac{4}{6}\right) = \frac{15 + 6 - 12}{6} = \frac{9}{6} = \mathbf{+\1.50} Since $E(X) > 0$, the game is favorable to the player with an average expected profit of $x1.50$} per roll.

**17** **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 3.01 Advanced Functions**

A fair coin is tossed 6 times. What is the probability of obtaining exactly 4 heads?

Apply the binomial probability formula $P(X = k) = \binom{n}{k} p^k (1 - p)^{n - k}$ with $n = 6$, $k = 4$, and $p = 1/2$: $P(X = 4) = \binom{6}{4} \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^2 = \binom{6}{2} \left(\frac{1}{2}\right)^6 \binom{6}{2} = \frac{6 \times 5}{2} = 15$ $P(X = 4) = 15 \times \frac{1}{64} = \mathbf{\frac{15}{64}} \approx 0.2344 \quad (23.44\%)$

**18** **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 4.01 Advanced Trigonometry**

In a class of 30 students, 18 play football (F), 14 play basketball (B), and 6 play both sports. How many students play neither sport?

Step 1 (Inclusion-Exclusion Principle): $|F \cup B| = |F| + |B| - |F \cap B| = 18 + 14 - 6 = 32 - 6 = 26$ Step 2 (Neither Sport): Subtract the union from the total class size: $\text{Neither} = 30 - |F \cup B| = 30 - 26 = \mathbf{4}$

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**19** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Pre-Calculus Concepts

The ACT Subject Test in Mathematics consists of 50 questions in 60 minutes. If a student spends an average of 55 seconds on each of the first 30 questions, how many seconds per question are available for the remaining 20 questions?

Time spent on first 30 questions $= 30 \times 55 = 1650$ seconds (27.5 minutes).

Total time available $= 60 \times 60 = 3600$ seconds. Remaining time $= 3600 - 1650 = 1950$ seconds. For the final 20 questions: $\frac{1950}{20} = 97.5$ seconds per question.

**20** [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.01 Advanced Coordinate Geometry

A dataset of 40 exam scores has mean $\bar{x} = 72$ and sample standard deviation $s = 8$. The instructor rescales all scores using $y_i = 1.25x_i - 10$. What are the mean and standard deviation of the rescaled scores?

Step 1: Compute transformed mean: For a linear transformation $y = ax + b$, the mean transforms identically: $\bar{y} = 1.25\bar{x} - 10 = 1.25(72) - 10 = 90 - 10 = \mathbf{80}$. Step 2: Compute transformed standard deviation: The additive constant $-10$ shifts the entire distribution without changing spread. Only the multiplicative factor scales dispersion: $s_y = |a| \cdot s_x = 1.25 \times 8 = \mathbf{10}$.

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## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A charity raffle sells $1,000$ tickets at $5$ each. There is one grand prize of $1,000$ and two second prizes of $500$ each. What is the expected net gain for a person purchasing one ticket?

**a.** **-3.00** ✓

**b.** -2.00

**c.** -1.50

**d.** 0.00

Expected payout $= \frac{1}{1000}(1000) + \frac{2}{1000}(500) = 1 + 1 = 2$. Net expected gain $= 2 - 5 = -3$ dollars.

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**22** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A random variable $X$ has probability distribution: $P(X = 1) = 0.2, P(X = 2) = 0.5, P(X = 3) = 0.3$. What is the expected value $E(X)$?

**a.** 2.0

**b.** 2.1 ✓

**c.** 2.2

**d.** 2.3

$E(X) = 1(0.2) + 2(0.5) + 3(0.3) = 0.2 + 1.0 + 0.9 = 2.1$.

**23** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $E(X) = 4$ and $E(X^2) = 25$, what is the variance $\text{Var}(X)$?

**a.** 3

**b.** 5

**c.** 9 ✓

**d.** 16

$\text{Var}(X) = E(X^2) - [E(X)]^2 = 25 - 4^2 = 25 - 16 = 9$.

**24** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The scores on a standardized exam are normally distributed with a mean of $500$ and a standard deviation of $100$. Approximately what percentage of test takers score between $400$ and $600$?

**a.** 50%

**b.** 68% ✓

**c.** 95%

**d.** 99.7%

Scores between $400$ and $600$ lie within $1$ standard deviation of the mean ($\mu \pm \sigma$). By the Empirical Rule, this encompasses approximately $68\%$ of data.

**25** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A student scores $84$ on a test with mean $72$ and standard deviation $8$. What is the student's $z$-score?

**a.** 1.25

**b.** 1.50 ✓

**c.** 1.75

**d.** 2.00

Formula: $z = \frac{x - \mu}{\sigma} = \frac{84 - 72}{8} = \frac{12}{8} = 1.50$.

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**26** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In a normal distribution with mean $50$ and standard deviation $5$, what value corresponds to a $z$-score of $-1.6$?

**a.** $42$ ✓

**b.** $44$

**c.** $46$

**d.** $48$

$x = \mu + z\sigma = 50 + (-1.6)(5) = 50 - 8 = 42$.

**27** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Given $P(A) = 0.6, P(B) = 0.5,$ and $P(A \cap B) = 0.3$, what is the conditional probability $P(A \mid B)$?

**a.** $0.50$

**b.** $0.60$ ✓

**c.** $0.75$

**d.** $0.30$

$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{0.3}{0.5} = 0.60$.

**28** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If two events $A$ and $B$ are mutually exclusive with $P(A) = 0.35$ and $P(B) = 0.45$, what is $P(A \cup B)$?

**a.** $0.70$

**b.** $0.80$ ✓

**c.** $0.90$

**d.** $0.1575$

For mutually exclusive events, $P(A \cap B) = 0$. Therefore, $P(A \cup B) = P(A) + P(B) = 0.35 + 0.45 = 0.80$.

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A medical screening test is $95\%$ accurate. If $2\%$ of a population has the disease, what is the probability that a randomly chosen person has the disease AND tests positive?

**a.** $0.019$ ✓

**b.** $0.020$

**c.** $0.095$

**d.** $0.95$

$P(\text{Disease} \cap \text{Positive}) = P(\text{Disease}) \times P(\text{Positive} \mid \text{Disease}) = 0.02 \times 0.95 = 0.019$.

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**30** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In how many ways can 6 students be seated in a circle?

**a.** 120 ✓

**b.** 720

**c.** 60

**d.** 24

Circular permutations of $n$ distinct objects is $(n-1)!$. Here $(6-1)! = 5! = 120$.

**31** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the minimum number of socks that must be drawn from a drawer containing 10 black socks and 10 white socks to guarantee getting at least one matching pair?

**a.** 2

**b.** 3 ✓

**c.** 4

**d.** 11

There are 2 colors (pigeonholes). By the Pigeonhole Principle, drawing $2 + 1 = 3$ socks guarantees at least two socks of the same color.

**32** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

How many distinct permutations can be formed using all the letters of the word 'STATISTICS'?

**a.** 50,400 ✓

**b.** 100,800

**c.** 302,400

**d.** 3,628,800

Word length is 10 letters: 3 S's, 3 T's, 2 I's, 1 A, 1 C. Permutations $= \frac{10!}{3!\,3!\,2!} = \frac{3,628,800}{6 \times 6 \times 2} = \frac{3,628,800}{72} = 50,400$.

**33** [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A bag has 13 marbles, 2 of which are red. What is the probability of drawing a red marble?

**a.** $\frac{11}{13}$

**b.** $\frac{2}{13}$ ✓

**c.** $\frac{1}{13}$

**d.** $\frac{2}{11}$

$P(\text{red}) = \frac{2}{13} = \frac{2}{13}$.

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**34** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

How many ways can you choose 5 items from 13 distinct items, where order does not matter?

**a.** 1286

**b.** 1288

**c.** 154440

**d.** **1287** ✓

$C(13,5) = \frac{13!}{5!\,(13-5)!} = 1287.$

**35** [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

How many ways can you arrange 2 items chosen from 10 distinct items, where order matters?

**a.** 89

**b.** **90** ✓

**c.** 91

**d.** 45

$P(10,2) = \frac{10!}{(10-2)!} = 90.$

**36** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A set of exam scores is normally distributed with a mean of $\mu = 74$ and a standard deviation of $\sigma = 8$. What is the $z$-score corresponding to an exam score of $58$?

**a.** $-2$ ✓

**b.** $-0.5$

**c.** $0.5$

**d.** $2$

To find the $z$-score corresponding to a given data value, use the standard normal distribution formula: $z = \frac{x-\mu}{\sigma}$ where $x$ is the individual score, $\mu$ is the population mean, and $\sigma$ is the standard deviation. Substitute the given values into the formula: $z = \frac{58-74}{8} = \frac{-16}{8} = -2$ A $z$-score of $-2$ indicates that the score of $58$ is located $2$ standard deviations below the mean. Common Misconceptions: - $2$: Results from subtracting the score from the mean.

**37** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The table shows exam marks for 20 students. Find the mean and identify the modal class.

**a.** **Mean ≈ 65, Modal class 60–69** ✓

**b.** Mean ≈ 68, Modal class 70–79

**c.** Mean ≈ 60, Modal class 50–59

**d.** Mean ≈ 72, Modal class 60–69

Mid-points: 44.5(2)+54.5(4)+64.5(7)+74.5(5)+84.5(2) = 89+218+451.5+372.5+169 = 1300; mean = 1300/20 = 65. Modal class = 60–69 (highest frequency).

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## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Advanced Trigonometry

A basketball player makes a free throw with probability $2/3$. If she takes $2$ free throws, what is the probability she makes at least one?

**a.** 53/90

**b.** 55/72

**c.** **8/9** ✓

**d.** 35/36

$P(\text{at least one}) = 1 - P(\text{none}) = 1 - (1/3)^2 = 1 - 1/9 = 8/9$.

**39** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Pre-Calculus Concepts

Given that $\sin\theta = \frac{5}{13}$ and $\frac{\pi}{2} < \theta < \pi$, what is the value of $\cos\theta$?

**a.** $-\frac{12}{13}$ ✓

**b.** $-\frac{5}{12}$

**c.** $\frac{12}{13}$

**d.** $\frac{13}{12}$

We use the fundamental Pythagorean trigonometric identity: $\sin^2\theta + \cos^2\theta = 1$ Substitute $\sin\theta = \frac{5}{13}$: $\left(\frac{5}{13}\right)^2 + \cos^2\theta = 1 \Rightarrow \frac{25}{169} + \cos^2\theta = 1$ $\cos^2\theta = 1 - \frac{25}{169} = \frac{144}{169}$ Taking the square root gives: $\cos\theta = \pm\sqrt{\frac{144}{169}} = \pm\frac{12}{13}$ Since the angle $\theta$ lies in Quadrant II ($\frac{\pi}{2} < \theta < \pi$), cosine is negative. Therefore, $\cos\theta = -\frac{12}{13}$. Common.

**40** [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.01 Advanced Coordinate Geometry

If $f(x) = 2x - 1$ and $g(x) = x^2 + 3$, what is the value of $g(f(2))$?

**a.** 9

**b.** **12** ✓

**c.** 13

**d.** 21

To find the composite value $g(f(2))$, evaluate from the inside out: 1. Evaluate the inner function at $x = 2$: $f(2) = 2(2) - 1 = 4 - 1 = 3$ 2. Substitute this result into the outer function $g(x)$: $g(3) = 3^2 + 3 = 9 + 3 = 12$ Common Misconceptions: - 13 results from reversing the order of operations to evaluate $f(g(2))$: $g(2) = 2^2 + 3 = 7$, and $f(7) = 2(7) - 1 = 13$. - 21 results from multiplying the two values instead of composing them: $f(2) \cdot g(2) = 3 \cdot 7 = 21$. - 9 results from squaring $f(2)$ b.

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**41** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Advanced Plane Geometry

If $\sin\theta = -\frac{20}{101}$ and $\theta$ is in Quadrant III, find $\cos\theta$.

**a.** $-\frac{99}{101}$ ✓

**b.** $-\frac{20}{101}$

**c.** $\frac{20}{101}$

**d.** $\frac{99}{101}$

By the Pythagorean identity, $\cos\theta = \pm\sqrt{1-\sin^2\theta} = \pm\frac{99}{101}$. In Quadrant III, the requested ratio is negative, so the value is $-\frac{99}{101}$.

**42** [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Advanced Number Theory

If $\cos\theta = \frac{99}{101}$ and $\theta$ is in Quadrant IV, find $\sin\theta$.

**a.** $\frac{20}{101}$

**b.** $\frac{99}{101}$

**c.** $-\frac{20}{101}$ ✓

**d.** $-\frac{99}{101}$

By the Pythagorean identity, $\sin\theta = \pm\sqrt{1-\cos^2\theta} = \pm\frac{20}{101}$. In Quadrant IV, the requested ratio is negative, so the value is $-\frac{20}{101}$.

**43** [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 2.01 Advanced Algebra & Systems

If $\cos\theta = -\frac{40}{41}$ and $\theta$ is in Quadrant II, find $\sin\theta$.

**a.** $\frac{9}{41}$ ✓

**b.** $\frac{40}{41}$

**c.** $-\frac{9}{41}$

**d.** $-\frac{40}{41}$

By the Pythagorean identity, $\sin\theta = \pm\sqrt{1-\cos^2\theta} = \pm\frac{9}{41}$. In Quadrant II, the requested ratio is positive, so the value is $\frac{9}{41}$.

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**44** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Advanced Functions

Find the vector projection of $\mathbf{u} = \langle 2, 2 \rangle$ onto $\mathbf{v} = \langle 3, 4 \rangle$.

**a.** $\frac{14}{25}\langle 3, 4 \rangle$ ✓

**b.** $\frac{14}{5}\langle 3, 4 \rangle$

**c.** $\frac{14}{25}\langle 2, 2 \rangle$

**d.** $\frac{19}{25}\langle 3, 4 \rangle$

$\text{proj}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{v}||^2}\mathbf{v} = \frac{14}{25}\langle 3, 4 \rangle$.

**45** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Advanced Trigonometry

If $\sin\theta = \frac{1}{3}$ in Quadrant I, find the exact value of $\sin(2\theta)$.

**a.** $7/9$

**b.** $4\sqrt{2}/9$ ✓

**c.** $-7/9$

**d.** $-4\sqrt{2}/9$

Using $\cos\theta = 2\sqrt{2}/3$, $\sin(2\theta) = 2\sin\theta\cos\theta = 2\left(\frac{1}{3}\right)(2\sqrt{2}/3) = 4\sqrt{2}/9$.

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